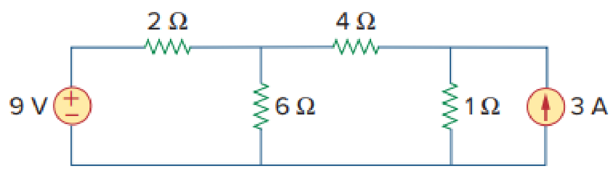
Chapter 8, Problem 74P

Problem

Draw the dual of the circuit shown in Fig. 8.118.

Figure 8.118:



Step-by-step solution

Step 1 of 5

Consider the following circuit diagram:

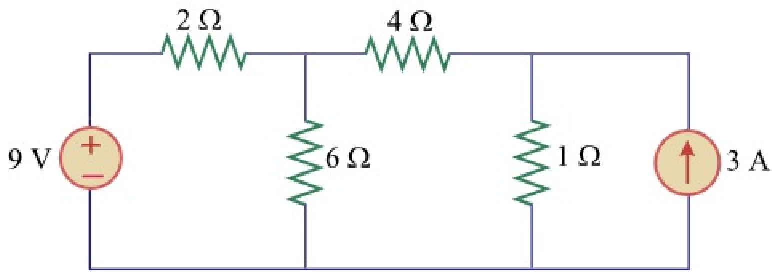


Figure 1

Step 2 of 5

Steps to follow to draw the dual of the circuit in Figure 1 are:

<u>*Step1:*</u>

First locate nodes at the center of the each mesh and a reference node as; node 1, 2, and 3 for three meshes and also the ground node 0 for the dual circuit as shown in Figure 2.

Redraw the circuit in Figure 1 by placing nodes:

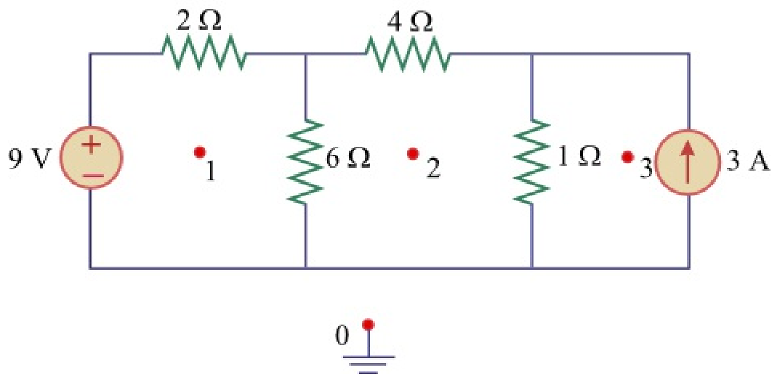


Figure 2

Step 3 of 5

Step2:

Draw the lines between the nodes as shown in Figure 3 such that each line crosses an element. Replace the element by its dual;

o Replace a line between the nodes 1 and 2 crossing the $6\ \Omega$ resistor by its dual $0.667\ ℧$ conductor.

o Replace a line between the nodes 1 and 0 crossing the 9 V voltage source by its dual 9 A current source whose reference direction is from ground to the nonreference node 1 and $2\ \Omega$ resistor by its dual $0.5\ ℧$ conductor.

o Replace a line between the nodes 2 and 0 crossing the $4\ \Omega$ resistor by its dual $0.25\ ℧$ conductor.

o Replace a line between the nodes 2 and 3 crossing the $1\ \Omega$ resistor by its dual $1\ ℧$ conductor.

o Replace a line between the nodes 3 and 0 crossing the $3\ \text{A}$ current source by its dual $3\ \text{V}$ voltage source.

Step 4 of 5

The following is the construction of the dual circuit of Figure 2.

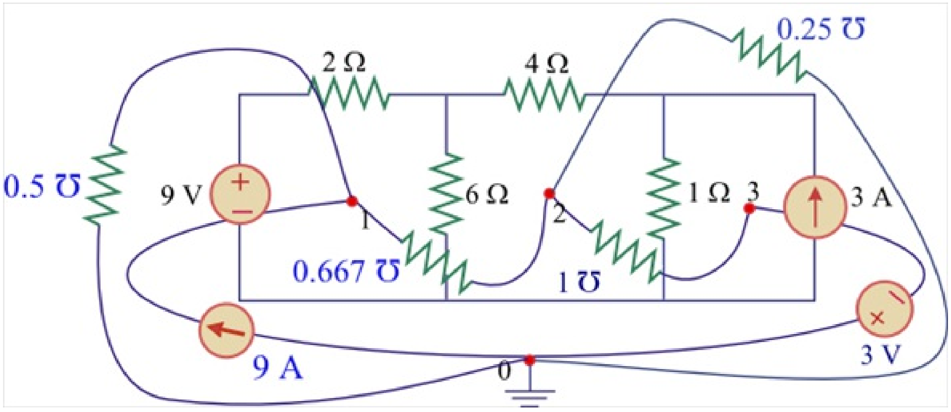


Figure 3

Redraw the circuit in Figure 3.

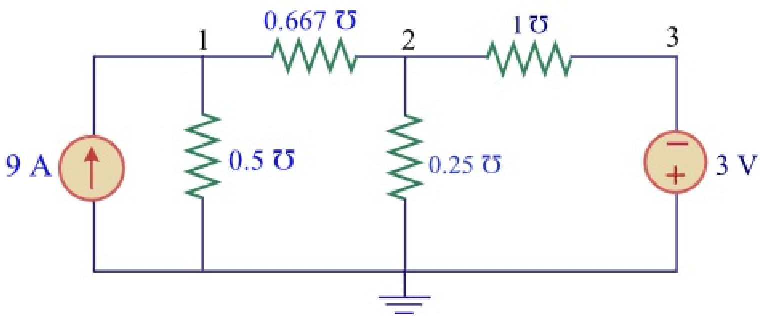


Figure 4: dual circuit for Figure 1

Step 5 of 5

Thus, the circuit in Figure 4 is the dual of the circuit in Figure 1.